

Multiple Choice (10 marks)

1. The AH Protocol provides source authentication and data integrity, but not
 - (a) Integrity
 - (b) Privacy
 - (c) Nonrepudiation
 - (d) Both A & C
2. The digest created by a hash function is normally called a
 - (a) Modication detection code (MDC)
 - (b) Modify authentication connection
 - (c) Message authentication control
 - (d) Message authentication cipher
3. Which of the following deals with network intrusion detection and real-time traffic analysis?
 - (a) John the Ripper
 - (b) L₀ phtCrack
 - (c) Snort
 - (d) Nessus
4. What was the first buffer overflow attack?
 - (a) Love Bug
 - (b) SQL Slammer
 - (c) Morris Worm
 - (d) Code Red
5. _____ is alike as that of Access Point (AP) from 802.11, & the mobile operators uses it for offering signal coverage.
 - (a) Base Signal Station
 - (b) Base Transmitter Station
 - (c) Base Transceiver Station
 - (d) Transceiver Station

True / False (10 marks)

Answer True or False

6. True or False: The authentication process between a wireless user and an access point involves a four-step procedure known as the 4-way handshake.
7. True or False: EtterPeak is a widely used tool for network analysis in multiprotocol diverse networks.
8. True or False: Eavesdropping is considered a security exploit that can compromise sensitive information during data transmission.
9. True or False: The Tor browser is essential for accessing the Tor network and utilizing its various services while maintaining anonymity.
10. True or False: A SYN stealth scan, also known as a half-open scan, does not complete the TCP handshake to identify open ports.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the semantic security of the encryption scheme defined by $E(k,m) = [r \leftarrow R, \text{output } (r, F(k,r) \oplus m)]$ under chosen plaintext attacks, considering the implications of the randomness of r .
12. Discuss the various issues related to the presentation layer in cybersecurity, specifically analyzing the impact of weak or non-existent authentication mechanisms on system security.

End of Paper